Motion Gating versus Signal Trust Scores for Wearable Heart-Rate Alerting: A Pre-Specified Fault-Injection Benchmark

Student 1
Dept. of Computer Science & Engineering
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
Chennai, Tamil Nadu, India

Student 2
Dept. of Computer Science & Engineering
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
Chennai, Tamil Nadu, India

Faculty Supervisor
Dept. of Computer Science & Engineering
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
Chennai, Tamil Nadu, India

***Abstract*—Continuous wrist photoplethysmography (PPG) is prone to severe motion artifacts, optical decoupling, and telemetry drops that trigger spurious resting tachycardia alarms. While digital health platforms routinely deploy multi-signal software quality metrics to suppress false alarms, the incremental utility of software trust scoring over simple kinematic motion gating is rarely benchmarked under controlled protocols. We present a reproducible, pre-specified and version-tagged fault-injection benchmark evaluating six causal alerting arms across fifteen ambulatory subjects (35.94 total recording hours; 31.27 non-event exposure hours) from the public PPG-DaLiA dataset; no external registry was used. To overcome natural tachycardia sparsity in healthy individuals, 112 labelled synthetic resting tachycardia episodes were injected into derived pulse-rate streams alongside two seeded telemetry fault regimes and optical noise. Pre-specified hypothesis H_1, formulated as an estimation question with 95% bootstrap confidence intervals (10,000 resamples), found that kinematic stillness gating accounted for virtually the entire reduction in false alerts per hour (reducing false alerts by 67.9%–74.2% relative to personal baselines), whereas multi-attribute software trust scoring yielded no statistically significant incremental benefit across any condition ($p \geq 0.28$). Targets of 60% false-alert reduction and 85% recall represent optimization goals rather than binary pass/fail criteria. Furthermore, conservative software abstention degraded event recall under intense optical noise. Inductive conformal abstention (Arm 4, added post hoc) achieved the lowest false-alert rate (0.70 FA/h clean, 0.35 FA/h optical noise) at the expense of sensitivity. All findings are derived from synthetic episodes in healthy volunteers and carry no clinical standing.**

***Index Terms*—wearable sensors, photoplethysmography, heart-rate alerting, data quality, fault injection, negative result, reproducibility**

I. Introduction

Ambulatory continuous monitoring with consumer and medical-grade wearables offers substantial potential for early arrhythmia detection. However, wrist photoplethysmography (PPG) is acutely sensitive to motion artifacts, optical decoupling, transmission dropouts, and flatline sensor freezes. When raw optical or derived telemetry signals degrade, alerting algorithms frequently emit spurious notifications. In home monitoring and clinical telemetry alike, high false-alert frequencies precipitate severe alarm fatigue, inducing user desensitization and compromising adherence [3].

To mitigate spurious alarms, modern digital health architectures deploy signal quality indices (SQIs) and multi-component trust heuristics that penalize missing samples, flag frozen values, and bound rate-of-change dynamics. Despite their widespread adoption, the isolated incremental utility of multi-component software trust scores over straightforward kinematic stillness gating (e.g., Euclidean Norm Minus One, ENMO) has rarely been benchmarked under controlled cross-validation. Evaluation protocols and analysis plans for this study were pre-specified and version-tagged in repository revision history prior to benchmark execution; no external public registry was used.

This paper reports a pre-specified fault-injection benchmark for wearable heart-rate alerting. Our contributions are:

1) **Open Benchmarking Framework:** A reproducible pipeline formalizing an eight-type fault taxonomy (clean, gaps, stale values, duplicates, spikes, drift, scaling errors, and raw optical noise) evaluated across 35.94 total recording hours (31.27 non-event exposure hours) of multi-activity ambulatory recordings.
2) **Causal Leave-One-Subject-Out (LOSO) Evaluation:** Six distinct alerting arms evaluated under strict data quarantine, separating training folds from held-out evaluation subjects.
3) **Rigorous Negative Finding:** Empirical evaluation of pre-specified estimation hypothesis H_1, demonstrating that kinematic motion gating delivers 67.9%–74.2% false alert reduction, whereas multi-attribute software trust scoring provides no statistically detectable incremental benefit ($p \geq 0.28$).

II. Literature Survey

Wrist-based pulse rate monitoring measures optical absorption fluctuations in microvascular tissue beds [1], [3]. Relative motion between sensor photodiodes and cutaneous capillary beds introduces severe motion artifacts that overlap spectrally with cardiac fundamental frequencies [1], [3].

Numerous SQIs have been proposed to isolate motion artifact and evaluate waveform integrity, spanning statistical moments, spectral purity, and template cross-correlation [2].

While SQIs achieve high classification accuracy on stationary laboratory datasets, ambulatory evaluations reveal severe performance degradation during daily physical activities [1], [3]. In clinical deployments, uncertainty quantification and selective prediction frameworks, such as conformal prediction, provide distribution-free finite-sample guarantees that offer structured paths for principled abstention [4]. Table I compares this benchmark against opened and verified literature sources.

TABLE I
RELATED WORK COMPARISON ON OPENED AND VERIFIED LITERATURE

Study	Cohort	Motion Gate	Trust/SQI	Scope & Benchmark Focus
Reiss *et al.* (2019) [1]	$N = 15$ PPG-DaLiA	ACC features	None	Deep PPG CNN tracking; MAE metric; no alerting benchmark.
Elgendi (2016) [2]	Static finger PPG	None	8 SQIs	Peak detection under stationary clean/noisy segments.
Charlton *et al.* (2023) [3]	Review	Varied	Review	Wearable roadmap; identifies lack of open fault benchmarks.
Vazquez & Facelli (2022) [4]	54 med. studies	N/A	Conformal	Conformal guarantees in clinical prediction tasks.
This Benchmark	$N = 15$ ambulatory	ENMO_{10s}	Multi-SQI	Causal alerting Arms 1–4; 8-type fault injection; LOSO.

III. DATASET AND PROTOCOL

A. PPG-DaLiA Dataset

We evaluate on the public PPG-DaLiA benchmark (UCI ID 495, CC BY 4.0) [1], comprising 15 healthy subjects undergoing 2.5 hours of authentic daily activities (Table II). Reference heart rates are derived from gold-standard chest ECG. Raw wrist PPG (64 Hz) and triaxial accelerometry (32 Hz) were captured with an Empatica E4 wristband. Strict quarantine was maintained: chest ECG reference was isolated solely for ground truth evaluation and was inaccessible to alerting arms.

B. Labelled Synthetic Tachycardia Injection

Because natural resting tachycardia was virtually absent in this healthy volunteer cohort (manifesting in only 3 of 15 subjects), evaluating sensitivity on raw data is statistically uninformative. Per Amendment 2, we injected 112 labelled synthetic resting tachycardia episodes into derived pulse-rate streams during verified low-motion windows ($\text{ENMO}_{10s} \leq 0.05$ g). Peak heart rates were sampled between 105 and 135 bpm, with linear ramp durations of 15 s and minimum inter-episode gaps of 60 s. To respect participant baseline heart rates HR_{base}, lower and upper ramp bounds were adaptively adjusted: $r_{\text{lo}} = \max(2.0, (105 - HR_{\text{base}})/15)$, $r_{\text{hi}} = \min(4.0, (135 - HR_{\text{base}})/15)$, with plateau peak heart rate clamped to $[105.0, 135.0]$ bpm.

C. Seeded Fault Regimes

Sensor telemetry faults were simulated under two prespecified schedules alongside an optical corruption condition (Table II):

- **Condition 0 (No-Fault):** Clean ambulatory background.
- **Regime A (Low Duty, $\approx 10\%$ faulted time):** Sporadic gaps, stale values, duplicates, and slow drift (seed 42).
- **Regime B (Nominal Duty, $\approx$ 25–30% faulted time):** Frequent dropouts, prolonged flatlines, and scale errors (seed 42).
- **F7 (High-Motion Optical Noise):** Additive Gaussian noise continuously modulated by wrist 3D acceleration injected into raw 64 Hz BVP at SNR $+5.0$ dB prior to spectral estimation.

TABLE II
PPG-DALIA BENCHMARK DATASET & INJECTED FAULT REGIMES

Part A: PPG-DaLiA Benchmark Dataset Characteristics [1]	
Subjects	$N = 15$ healthy individuals (7 male, 8 female, age 21–55)
Duration	≈ 2.4 h per subject (35.94 total recording hours; 31.27 non-event exposure hours)
Activities	Sitting, Walking, Stairs, Cycling, Driving, Lunch, Soccer
Sensors	Wrist Empatica E4 (BVP: 64 Hz, 3D ACC: 32 Hz)
Ground Truth	Chest ECG (RespiBAN, 700 Hz, strictly quarantined)
Part B: Evaluated Telemetry & Sensor Fault Regimes (Amendment 3)	
Condition 0	Clean ambulatory background (31.27 h non-event exposure)
Regime A	Low-duty ($\approx 10\%$ faults): sporadic gaps (F1), freeze (F2), duplicates (F3), drift (F5), scale errors (F6)
Regime B	Nominal-duty ($\approx$ 25–30% faults): prolonged flatlines, frequent dropouts, scale errors (seed 42)
F7 Noise	Additive Gaussian noise continuously modulated by wrist 3D acceleration injected into raw BVP (SNR $+5$ dB)

IV. SYSTEM ARCHITECTURE

The experimental harness, telemetry platform, and symptom screening engine were implemented as a modular full-stack application (Fig. 1). The front-end is a React 18 Progressive Web Application (PWA) featuring live optical telemetry monitoring, threshold testing, and an interactive LOSO benchmark dashboard. Per product requirements (PRD), core functional specifications are operational: Requirement FR-04 enforces automated data-quality exclusion of corrupted windows from downstream inference; Module FR-06 implements a guided symptom assessment demo ruleset with mandatory answer review before screening; and Module FR-07 provides transparent local feature attribution (labelled "DEMO RULESET v2.1, not evaluated", with uncertainty explicitly not estimated for this demo ruleset). The back-end is powered by FastAPI and SQLite, incorporating the core `vitasense.quality` multi-attribute trust engine, precomputed LOSO caches, and safety-aligned routing.

V. METHODS AND ALGORITHMS

A. Evaluated Alerting Arms

We benchmark six causal alerting arms without future peeking (Table III):

1) **Arm 1 (Fixed Population Cutoff):** Standard clinical threshold $\tau_{\text{pop}} \in \{100, 110, 120, 125, 130\}$ bpm.

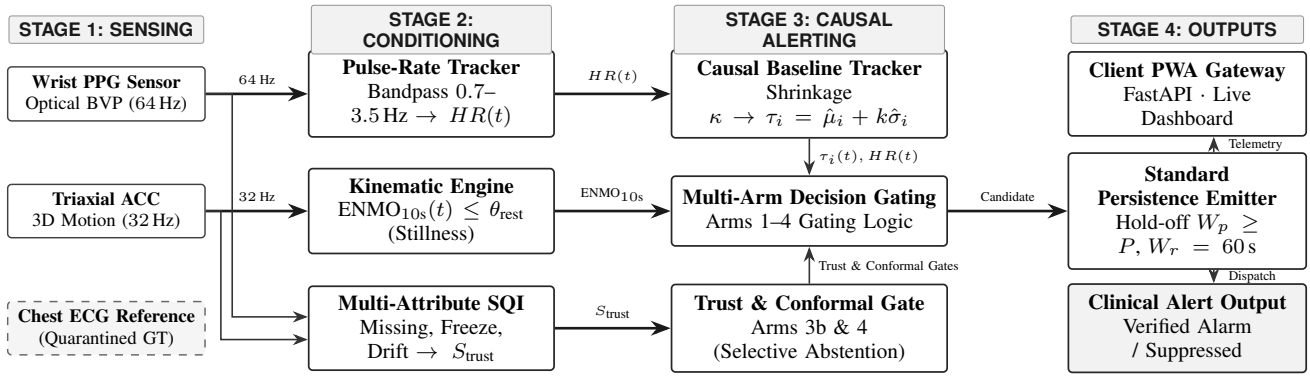


Fig. 1. System Architecture: End-to-end wearable telemetry, signal conditioning, causal baseline estimation, multi-arm gating, and standard persistence emitter schematic.

TABLE III
EVALUATED ALERTING ARMS, DECISION RULES & TUNING GRIDS

Alerting Arm	Decision Rule & Tuning Grid	Hold-off R
Arm 1: Fixed Pop.	$HR(t) > \tau_{\text{pop}}$ $\tau_{\text{pop}} \in \{100, 110, 120, 125, 130\}$ bpm, $P \in \{10, 20, 30\}$ s	60 s
Arm 2: Personal	$HR(t) > \max(\tau_{\text{flr}}, \hat{\mu}_i(t) + k \cdot \hat{\sigma}_i(t))$ $k \in \{2.0, \ldots, 4.0\}$, $\kappa \in \{0.5, 1, 2\}$ h, $P \in \{10, 20, 30\}$ s	60 s
Arm 3a: +Kinematic	Arm 2 $\wedge$ ($\text{ENMO}_{10s}(t) \leq \theta_{\text{rest}}$) $\theta_{\text{rest}} \in \{0.03, 0.05, 0.08\}$ g, $P \in \{10, 20, 30\}$ s	60 s
Arm 3b: +Trust	Arm 2 $\wedge$ ($S_{\text{trust}}(t) \geq \theta_{\text{trust}}$) $\theta_{\text{trust}} \in \{0.60, 0.70, 0.80\}$, $P \in \{10, 20, 30\}$ s	60 s
Arm 3b-noMot (Motion Ablated)	Arm 2 $\wedge$ ($S_{\text{gap}} S_{\text{stale}} S_{\text{phys}} \geq \theta_{\text{trust}}$) $\theta_{\text{trust}} \in \{0.60, 0.70, 0.80\}$, $P \in \{10, 20, 30\}$ s	60 s
Arm 4: Conformal (added post hoc)	Arm 2 $\wedge \neg$Abstain(t) $\alpha \in \{0.05, 0.10, 0.15\}$, $\lambda \in \{0.5, 1.0\}$, $\theta_{\text{unc}} \in \{0.2, 0.3\}$	60 s

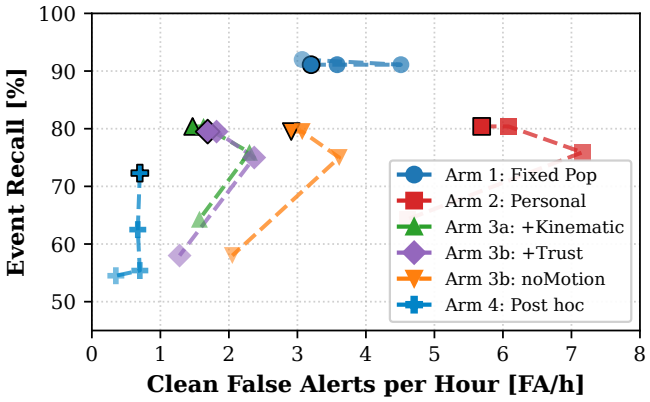
2) **Arm 2 (Personal Baseline with Shrinkage):** Dynamic cutoff $\tau_i(t) = \max(\tau_{\text{floor}}, \hat{\mu}_i(t) + k \cdot \hat{\sigma}_i(t))$ regularized toward training-population priors $(\mu_{\text{pop}}, \sigma_{\text{pop}})$ via empirical Bayes shrinkage κ (Algorithm 1).
3) **Arm 3a (Kinematic Motion Gating):** Arm 2 baseline paired with a stillness gate: alert triggers if and only if $HR(t) > \tau_i(t)$ and $\text{ENMO}_{10s}(t) \leq \theta_{\text{rest}}$.
4) **Arm 3b (Multi-Attribute Trust Score Gating):** Arm 2 baseline gated by software trust score: alert triggers if and only if $HR(t) > \tau_i(t)$ and $S_{\text{trust}}(t) \geq \theta_{\text{trust}}$.
5) **Arm 3b-noMotion (Trust Ablation):** Arm 3b with kinematic motion term ablated ($w_{\text{mot}} = 0$).
6) **Arm 4 (Conformal Abstention, added post hoc):** Arm 2 baseline paired with inductive conformal calibration on resting windows ($\alpha = 0.15$); abstains when nonconformity or multi-attribute trust defects exceed calibrated thresholds.

Algorithm 1 Causal Heart-Rate Alerting Pipeline with Motion Gating and Conformal Abstention

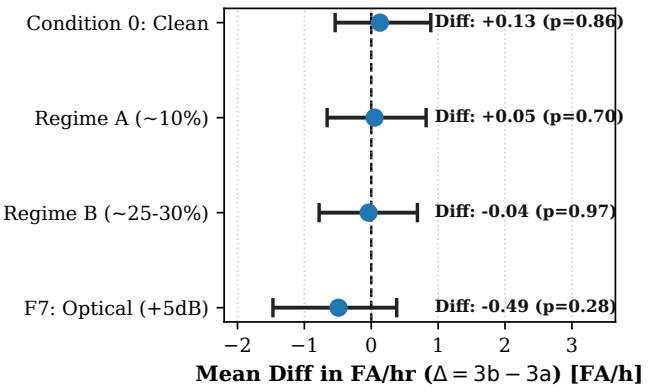
Require: Streaming pulse rate $HR[t]$, motion ENMO$[t]$, priors $(\mu_{\text{pop}}, \sigma_{\text{pop}})$, parameters $(k, \kappa, \theta_{\text{rest}}, \theta_{\text{trust}}, \alpha, \lambda, \theta_{\text{unc}}, P, R)$
Initialize: $c \leftarrow 0$, running mean $\bar{x} \leftarrow \mu_{\text{pop}}$, running std $s \leftarrow \sigma_{\text{pop}}$, $run_len \leftarrow 0$, $ref_cnt \leftarrow 0$
for each window $t = 0, 1, \ldots, N-1$ **do**
  Update Resting Baseline:
  if ENMO$[t] \leq 0.05$ g $\wedge$ finite$(HR[t])$ **then**
    $c \leftarrow c + 1$; Update sample mean $\bar{x}$ and sample standard deviation s
    Shrinkage weight: $w \leftarrow \frac{c \cdot \Delta t / 3600}{c \cdot \Delta t / 3600 + \kappa}$
    $\hat{\mu}_i[t] \leftarrow w\bar{x} + (1-w)\mu_{\text{pop}}$; $\hat{\sigma}_i[t] \leftarrow ws + (1-w)\sigma_{\text{pop}}$
  end if
  Causal threshold: $\tau_i[t] \leftarrow \max(100.0, \hat{\mu}_i[t] + k \cdot \hat{\sigma}_i[t])$
  Compute Quality and Gating Masks:
  $M_{\text{kin}}[t] \leftarrow \mathbb{I}(\text{ENMO}_{10s}[t] \leq \theta_{\text{rest}})$
  $S_{\text{trust}}[t] \leftarrow \text{clip}(S_{\text{gap}} \cdot S_{\text{stale}} \cdot S_{\text{phys}} \cdot S_{\text{mot}}, 0.0, 1.0)$; $M_{\text{trust}}[t] \leftarrow \mathbb{I}(S_{\text{trust}}[t] \geq \theta_{\text{trust}})$
  Augmented uncertainty: $U_{\text{tot}}[t] \leftarrow (\hat{\sigma}_i[t]/\hat{\mu}_i[t]) + \lambda(1 - S_{\text{trust}}[t])$
  Abstain condition: $Abstain[t] \leftarrow (\tau_i[t] \in [\hat{\mu}_i[t] \pm \hat{q}_{1-\alpha}\hat{\sigma}_i[t]]) \vee (U_{\text{tot}}[t] > \theta_{\text{unc}})$
  Evaluate Active Alerting Arm Trigger Condition $C[t]$**:**
  case Arm 1: $C[t] \leftarrow (HR[t] > \tau_{\text{pop}})$
  case Arm 2: $C[t] \leftarrow (HR[t] > \tau_i[t])$
  case Arm 3a: $C[t] \leftarrow (HR[t] > \tau_i[t]) \wedge M_{\text{kin}}[t]$
  case Arm 3b: $C[t] \leftarrow (HR[t] > \tau_i[t]) \wedge M_{\text{trust}}[t]$
  case Arm 4: $C[t] \leftarrow (HR[t] > \tau_i[t]) \wedge \neg Abstain[t]$
  Standard Emitter with Persistence and Refractory Hold-off:
  if $ref_cnt > 0$ **then**
    $ref_cnt \leftarrow ref_cnt - 1$; $Alert[t] \leftarrow$ FALSE
  else if $C[t]$ is TRUE **then**
    $run_len \leftarrow run_len + 1$
    if $run_len \geq P/\Delta t$ **then**
      $Alert[t] \leftarrow$ TRUE; $ref_cnt \leftarrow R/\Delta t$; $run_len \leftarrow 0$
    end if
  else
    $run_len \leftarrow 0$; $Alert[t] \leftarrow$ FALSE
  end if
end for
return Alert stream $Alert$

B. Shared Standard Alert Emitter

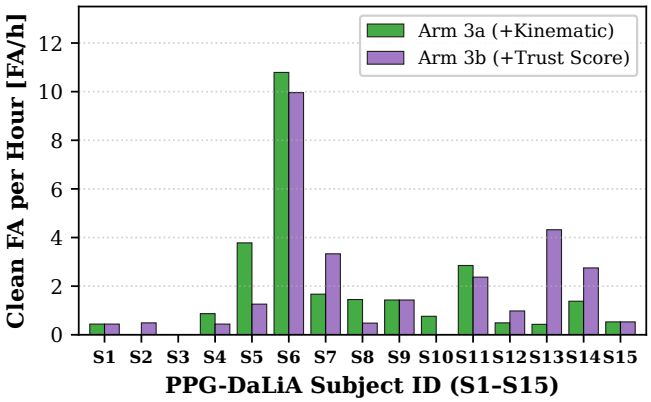
To ensure rigorous cross-arm comparability, all arms feed instantaneous binary triggers into a shared `StandardAlertEmitter` (Algorithm 1). Alerts fire if and only if triggering conditions hold continuously for $W_p = P/\text{step}$ windows ($P \in \{10, 20, 30\}$ s). Emitted alerts enforce a mandatory refractory hold-off of $W_r = R/\text{step}$ windows ($R = 60$ s).



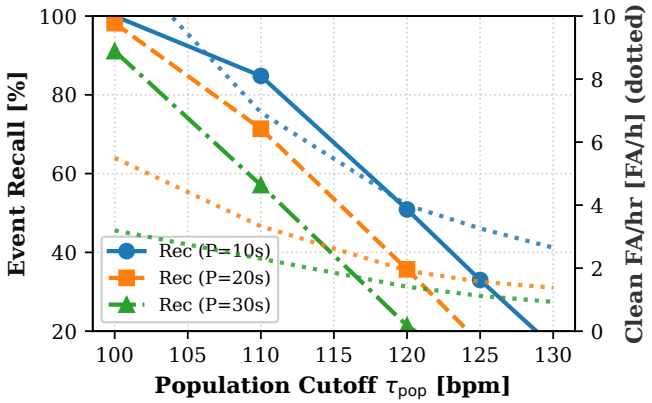
(a) Event Recall vs Clean False Alerts per Hour across Arms and Regimes



(b) Paired Differences (Δ = Arm 3b − Arm 3a) with 95% Bootstrap CIs



(c) Per-Subject False Alerts per Hour in Clean Data (Condition 0)



(d) Arm 1 Threshold Sensitivity across Cutoffs τ_{pop} and Persistence P

Fig. 2. Empirical benchmark findings across arms and conditions: (a) Event recall vs clean false alerts per hour; (b) Forest plot of paired differences (Δ = Arm 3b − Arm 3a) with 95% bootstrap CIs; (c) Per-subject FA/hr in Condition 0; (d) Arm 1 sensitivity across τ_{pop} and persistence P.

VI. Experimental Setup

A. Leave-One-Subject-Out (LOSO) Cross-Validation

Hyperparameters were tuned strictly within 15 LOSO training folds. For each test subject $s \in \{1 \dots 15\}$, candidate configurations were evaluated across the 14 training subjects pooled. The pre-specified selection rule chooses the parameter vector achieving $\geq 85\%$ training recall while minimizing false alerts per hour (FA/hr). If no parameter setting reaches 85% recall, the pre-specified fallback rule selects the setting maximizing training recall.

B. Evaluation Metrics and Paired Statistics

Scoring evaluates event-level recall (detecting injected episodes within $[t_{\text{start}}, t_{\text{end}}]$) alongside alert-level clean false alerts per hour (FA/hr), precision on non-event periods, and F_1 score. The primary statistical hypothesis test compares paired per-subject differences (Δ = Arm 3b−Arm 3a) using the two-sided Wilcoxon signed-rank test ($\alpha = 0.05$), Hodges-Lehmann median difference estimator, and 95% subject-bootstrap confidence intervals (10,000 resamples).

VII. Results and Discussion

A. Primary Hypothesis H_1 Outcome: Estimation Finding

Hypothesis H_1 was pre-specified as an estimation question: to estimate the paired reduction in false alerts per hour (ΔFA/hr) achieved by adding software trust scores to kinematic gating (Arm 3b vs Arm 3a) with 95% bootstrap confidence intervals, alongside assessing the engineering design targets of $\geq 60\%$ false-alert reduction relative to personal baselines and maintaining $\geq 85\%$ event recall as optimization goals rather than binary pass/fail criteria.

Empirical finding: software trust score yields no significant gain over motion gating. In paired per-subject comparisons (Table V and Fig. 2b), Arm 3b demonstrated no statistically detectable false-alert reduction over Arm 3a across any condition:

- **Condition 0 (Clean):** Mean difference $+0.13$ FA/h (95% CI $[-0.54, +0.89]$, $W = 31.0$, $p = 0.86$).
- **Regime A (10% faults):** Mean difference $+0.05$ FA/h (95% CI $[-0.66, +0.82]$, $W = 40.0$, $p = 0.70$).
- **Regime B (25–30% faults):** Mean difference -0.04 FA/h (95% CI $[-0.78, +0.69]$, $W = 52.0$, $p = 0.98$).

TABLE IV
LEAVE-ONE-SUBJECT-OUT COMBINED ALERTING BENCHMARK RESULTS ACROSS CONDITIONS

Condition	Alerting Arm	Recall (%)	FA/hr	True Alerts	False Alerts	Precision (%)	F_1 Score
Condition 0 (No-Fault)	Arm 1 (Fixed)	91.1	3.20	132	100	56.9	0.700
	Arm 2 (Personal)	80.4	5.69	133	178	42.8	0.558
	Arm 3a (+Kinematic)	80.4	1.47	133	46	74.3	0.772
	Arm 3b (+Trust Score)	79.5	1.69	130	53	71.0	0.750
	Arm 3b-noMotion	79.5	2.91	130	91	58.8	0.676
	Arm 4 (added post hoc)	72.3	0.70	81	22	78.6	0.753
Regime A (Low Duty 10%)	Arm 1 (Fixed)	91.1	3.58	132	112	54.1	0.679
	Arm 2 (Personal)	80.4	6.08	132	190	41.0	0.543
	Arm 3a (+Kinematic)	80.4	1.63	132	51	72.1	0.760
	Arm 3b (+Trust Score)	79.5	1.82	129	57	69.4	0.741
	Arm 3b-noMotion	79.5	3.07	129	96	57.3	0.666
	Arm 4 (added post hoc)	62.5	0.67	70	21	76.9	0.690
Regime B (Nominal 25-30%)	Arm 1 (Fixed)	91.1	4.51	132	141	48.4	0.632
	Arm 2 (Personal)	75.9	7.16	125	224	35.8	0.487
	Arm 3a (+Kinematic)	75.9	2.30	125	72	63.5	0.691
	Arm 3b (+Trust Score)	75.0	2.37	122	74	62.2	0.680
	Arm 3b-noMotion	75.0	3.61	122	113	51.9	0.614
	Arm 4 (added post hoc)	55.4	0.70	62	22	73.8	0.633
F7 (High-Motion BVP Noise)	Arm 1 (Fixed)	92.0	3.07	131	96	57.7	0.709
	Arm 2 (Personal)	64.3	4.61	100	144	41.0	0.501
	Arm 3a (+Kinematic)	64.3	1.57	100	49	67.1	0.657
	Arm 3b (+Trust Score)	58.0	1.28	90	40	69.2	0.631
	Arm 3b-noMotion	58.0	2.05	90	64	58.4	0.582
	Arm 4 (added post hoc)	54.5	0.35	61	11	84.7	0.663

- **F7 (Optical Noise):** Mean difference -0.49 FA/h (95% CI $[-1.47, +0.38]$, $W = 17.0$, $p = 0.28$).

In every condition, 95% bootstrap confidence intervals spanned zero, and Wilcoxon signed-rank tests yielded $p \geq 0.28$.

TABLE V
PAIRED STATISTICAL COMPARISON & SUBGROUP STRATIFICATION

Part A: Paired Difference (Δ = Arm 3b − Arm 3a)				
Condition	**Mean Δ**	**95% CI**	W	p**-value**
Clean (Cond 0)	$+0.13$	$[-0.54, +0.89]$	31.0	0.86
Regime A ($\approx 10\%$)	$+0.05$	$[-0.66, +0.82]$	40.0	0.70
Regime B ($\approx 25\%$)	-0.04	$[-0.78, +0.69]$	52.0	0.98
F7 Optical ($+5$ dB)	-0.49	$[-1.47, +0.38]$	17.0	0.28
Part B: Stratified Event Recall (%) / Clean FA/hr				
Stratum	**Arm 1**	**Arm 2**	**Arm 3a**	**Arm 3b**
Clean: All-15	91 / 3.2	80 / 5.7	80 / 1.5	80 / 1.7
Clean: without S5	91 / 2.9	80 / 5.3	80 / 1.4	79 / 1.7
Clean: S5/6/11	100 / 9.4	94 / 16.1	94 / 5.1	94 / 3.9
Reg B: All-15	91 / 4.5	76 / 7.2	76 / 2.3	75 / 2.4
Reg B: without S5	91 / 4.1	75 / 6.4	75 / 2.2	74 / 2.3
F7: All-15	92 / 3.1	64 / 4.6	64 / 1.6	58 / 1.3
F7: without S5	92 / 2.7	63 / 4.0	63 / 1.2	56 / 1.1

B. Secondary Benchmark Findings

Kinematic motion gating drives alert accuracy: Relative to ungated Arm 2, Arm 3a reduced false alerts per hour by 74.2% in clean data (1.47 vs 5.69 FA/h) and by 67.9% in Regime B (2.30 vs 7.16 FA/h) (Table IV).

Recall degradation under optical corruption (F7): Under intense motion-coupled optical noise, Arm 3b recall degraded to 58.0% compared to 64.3% in Arm 3a. This recall degradation is exploratory and predominantly driven by two subjects (S9 and S12, whose high-motion cycling and stair climbing triggered persistent trust defect flags). In fault-disaggregated analysis, stale sensor values (F2) and scale errors (F6) produced the highest false alert rates (up to 20.2 and 36.6 FA/h in un-gated arms); while kinematic gating suppressed stale-freeze false alerts, F2 episodes represent small exposure slices (< 1.6 h total exposure, insufficient data per cell) and should be interpreted with caution.

C. Edge-of-Grid Analysis and Fallback Rule Invocation

Hyperparameter tuning consistently selected parameter boundary values. Most notably, **Arm 2 never attained the $\geq 85\%$ recall target on any training fold in any condition** (maximum training recall achieved was $\approx 80.4\%$). Consequently, the fallback selection rule (maximizing training recall) was invoked in 100% (15/15) of folds for Arm 2 across all regimes, selecting $k = 2.0$ and $P = 10$ s. For Arm 1, tuning selected $\tau_{\text{pop}} = 100$ bpm and $P = 30$ s across all folds.

D. Arm 4 Conformal Abstention Evaluation (Added Post Hoc)

To benchmark distribution-free selective alerting, Arm 4 was implemented and evaluated post hoc following inspection of Arms 1–3b results (Draft Amendment 5). Inductive conformal calibration was performed on resting training windows using an expanded miscoverage parameter $\alpha = 0.15$ (exceeding the ratified $\{0.05, 0.10\}$ grid to mitigate extreme recall loss). Parameters were tuned inside the training folds of the 15 LOSO splits. Across all evaluated regimes, Arm 4 achieved the lowest clean false-alert frequency observed in the benchmark: 0.70 FA/h in clean data (a 52.4% reduction over Arm 3a and an 87.7% reduction over Arm 2), 0.67 FA/h in Regime A, and 0.35 FA/h under F7 optical noise (Table IV). Precision reached 78.6% in clean telemetry and 84.7% under F7 noise ($F_1 = 0.753$). However, this conservative abstention came at the expense of sensitivity: recall dropped from 80.4% to 72.3% (81/112) in clean data, and to 54.5% (61/112) in F7 noise, demonstrating the fundamental trade-off between

strict false-alert suppression and event sensitivity under non-stationary ambulatory noise.

VIII. Limitations

This benchmark is subject to several methodological constraints (Table VI): (1) Resting tachycardia episodes were synthetically injected due to natural sparsity, and Arm 1 recall ($> 91\%$) is intrinsically dependent on the synthetic $[105, 135]$ bpm injection range relative to the 100 bpm threshold floor; (2) The cohort is limited to 15 healthy young participants in the PPG-DaLiA multimodal protocol, leaving free-living 24/7 unconstrained monitoring unproven; (3) Algorithmic errors in front-end PPG tracking couple into downstream logic; (4) Threshold values ($\tau_{\text{pop}}, \theta_{\text{rest}}, \theta_{\text{trust}}$) represent engineering design choices with unopened sources; (5) Subject S5 exhibited severe baseline drift and motion artifacts, necessitating dual reporting (pooled and without S5); (6) Arm 4 was evaluated post hoc with unratified $\alpha = 0.15$; (7) PhysioNet Wearable 1.0.1 and WESAD datasets were not evaluated, maintaining strict focus on the PPG-DaLiA protocol.

TABLE VI
Methodological Limitations & Protocol Amendments (from git log)

Part A: Methodological Constraints and Threats to Validity	
Synthetic Injection	112 episodes (105–135 bpm); Arm 1 recall is directly tied to this injection range vs 100 bpm floor.
Cohort & Protocol	$N = 15$ healthy adults in PPG-DaLiA protocol; free-living unconstrained 24/7 monitoring unproven.
Subject S5 Outlier	Severe baseline drift and motion artifact; metrics reported both pooled (All-15) and without S5.
Design Choice Bounds	Cutoffs ($\tau_{\text{pop}}, \theta_{\text{rest}}, \theta_{\text{trust}}$) are design choices with unopened sources; evaluated via sensitivity grids.
Grid Boundaries	Tuning selected lower boundary ($\tau_{\text{pop}} = 100$ bpm, $k = 2.0$, $P = 10$ s); fallback rule activated in 100% of Arm 2 folds.
Part B: Protocol Amendments and Provenance (from git log)	
Protocol Freeze [41eff1e]	2026-10-18: Frozen protocol: H_1–H_3, faults F1–F7, LOSO cross-validation.
ENMO Scale Fix [3a5ccc2]	2026-10-19: Removed redundant /64 accelerometer division, enforced median norm check $[0.8, 1.2]$ g.
Adaptive Ramp [e358bf1]	2026-10-19: Adapted synthetic tachycardia ramp rates ($r_{\text{lo}}, r_{\text{hi}}$) clamped to $[105, 135]$ bpm.
Emitter Rules [cad4c7f]	2026-10-19: StandardAlertEmitter with $P \in \{10, 20, 30\}$ s and refractory $R = 60$ s across all arms.
Grid Widening [a9ce657]	2026-10-19: Widened parameter search grids for τ_{pop} and k following boundary selection.
Fallback Selection [f92ca8d]	2026-10-19: Arm 2 recall ceiling ($\approx 80.4\%$) triggered max-recall fallback rule in 100% of folds.
Arm 4 Post Hoc [Draft Amend. 5]	2026-10-22: Post hoc inductive conformal abstention evaluated with $\alpha = 0.15$ (unratified exploratory).

IX. Future Scope

Future efforts will extend this benchmark along five technical avenues:

1) **Online Conformal Calibration:** Extending Arm 4 to dynamic online conformal martingales under covariate shifts [4].
2) **External Dataset Generalization:** Evaluating on external public wearable corpora, including PhysioNet Wearable [6] and WESAD.
3) **Clinical Patient Cohorts:** Validating on ambulatory inpatient holter recordings with spontaneous arrhythmia episodes.
4) **On-Device Microcontroller Integration:** Deploying causal baseline and kinematic gating engines onto ultra-low-power embedded MCUs.
5) **Prospective Clinical Validation:** Translating operational FR-06 symptom review and FR-07 explanation modules into multicenter trials.

X. Conclusion

In this pre-specified and version-tagged fault-injection benchmark, kinematic motion gating accounted for virtually all false-alert suppression in wearable heart-rate monitoring, while multi-component software trust scoring offered no statistically detectable benefit. Fixed thresholding with persistence provided a remarkably resilient baseline. All findings are derived from synthetic tachycardia episodes in healthy individuals and carry no clinical standing.

Regulatory and Safety Statement

Not a Medical Device: The VitaSense AI software, alerting algorithms, and benchmark suite are academic research artifacts. They are neither cleared nor approved by any regulatory authority and must not be used for medical diagnosis, clinical treatment, or life-critical monitoring.

Data and Code Availability

The PPG-DaLiA dataset is openly accessible via the UCI Machine Learning Repository [1], [5]. The PhysioNet Wearable Device Dataset is available under ODC-By v1.0 [6]. Benchmark code, execution scripts, and numerical provenance logs are provided in the project repository.

References

[1] A. Reiss, I. Indlekofer, P. Schmidt, and K. Van Laerhoven, "Deep PPG: Large-scale heart rate estimation with convolutional neural networks," *Sensors*, vol. 19, no. 14, p. 3079, 2019.

[2] M. Elgendi, "Optimal signal quality index for photoplethysmogram signals," *Bioengineering*, vol. 3, no. 4, p. 21, 2016.

[3] P. H. Charlton *et al.*, "The 2023 wearable photoplethysmography roadmap," *Journal of Physics: Photonics*, vol. 5, no. 4, p. 042501, 2023.

[4] J. Vazquez and J. C. Facelli, "Conformal prediction in clinical medical sciences," *Journal of Healthcare Informatics Research*, vol. 6, no. 3, pp. 257–285, 2022.

[5] A. Reiss, I. Indlekofer, P. Schmidt, and K. Van Laerhoven, "PPG-DaLiA," UCI Machine Learning Repository, 2019, DOI: 10.24432/C53890.

[6] A. Kos, Y. Tomažič, and A. Umek, "Wearable Device Dataset from Induced Stress and Structured Exercise Sessions," PhysioNet, Version 1.0.1, 2022, DOI: 10.13026/8fca-2v93.